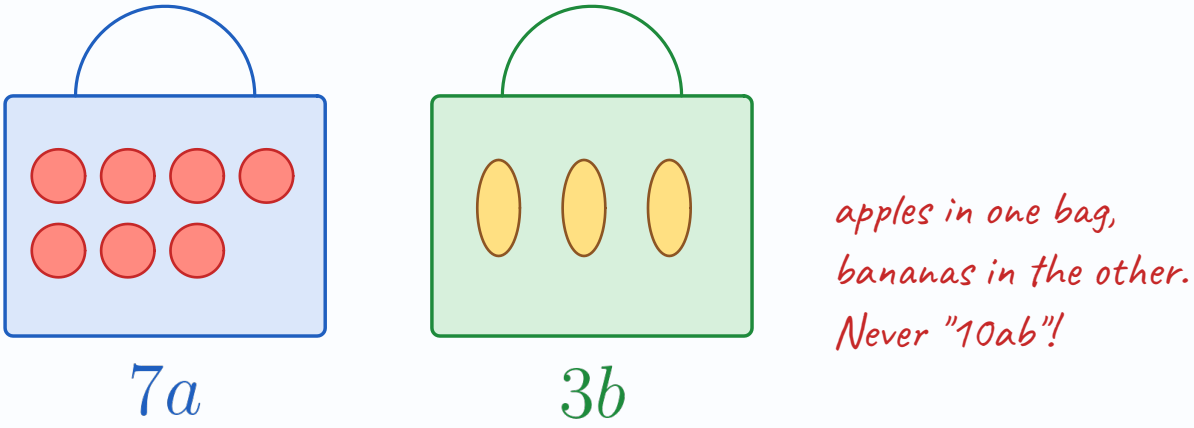
Chapter 11

Simplifying algebraic expressions

sorting the shopping, and multiplying out brackets with a grid



Simplifying means rewriting an expression so it's **shorter but equal**: the same value for every number you could put in. Two skills do almost all of the work: **collecting like terms** (sorting the shopping into bags) and **expanding brackets** (multiplying everything inside by what's outside).

Every rule is tested with numbers, so you'll never have to wonder "is this allowed?". That same habit catches two mistakes on the Lecture 2 slides, near the end of the chapter.

Goes with **Lecture 2** (expanding brackets, common mistakes).

Like terms: same fruit, same bag

RULE: collecting like terms

Like terms have exactly the **same letters** raised to the **same powers**. Only the number in front (the coefficient) may differ: $3x^2$ and $-5x^2$ are like; x and x^2 are not.

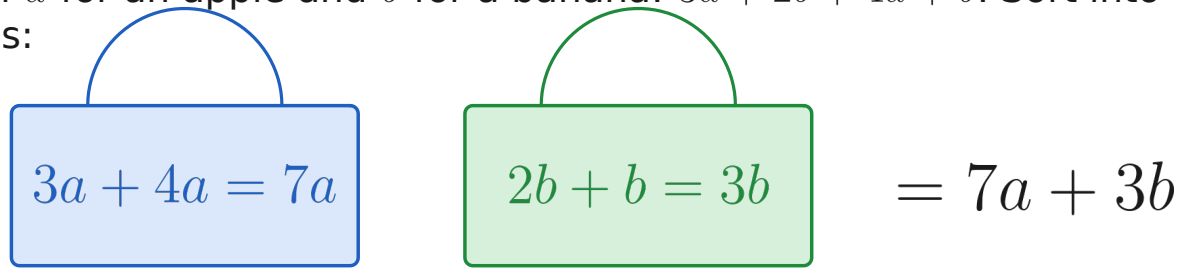
To collect like terms, add or subtract their **coefficients**; the letters and powers stay exactly as they are: $3x^2 - 5x^2 = -2x^2$.

Unlike terms cannot be combined by adding: $7a + 3b$ and $x + x^2$ are already as simple as they get.

The order of letters inside a term doesn't matter: xy and yx are like terms.

Terms	Like?	Why
$3x$ and $5x$	**yes**	same letter, same power: $3x + 5x = 8x$
x and x^2	**no**	same letter, different powers
$2xy$ and $5yx$	**yes**	yx is xy: $2xy + 5yx = 7xy$
$4a^2b$ and $3ab^2$	**no**	the powers sit on different letters
7 and -2	**yes**	both constants: $7 - 2 = 5$

Your shopping: **3 apples, 2 bananas, 4 apples, 1 banana**. Written with a for an apple and b for a banana: $3a + 2b + 4a + b$. Sort into bags:



Terms with **exactly the same letters and powers** are **like terms**: they can be added by adding their coefficients. Different letters stay separate: $7a + 3b$ can't be squashed into $10ab$ (10 apple-bananas?).

Powers count as a different fruit. x and x^2 are **not** like terms. Test with $x = 3$: $x + x^2 = 3 + 9 = 12$, but "$2x^2$" would be 18 and "$2x$" would be 6. So $x + x^2$ stays as it is.

A bigger example, colour-coded by kind:

$$5x^2 + 3x - 2x^2 + 7 - x + 1$$

purple = x^2 bag, blue = x bag, orange = plain numbers

$$= (5-2)x^2 + (3-1)x + (7+1) = 3x^2 + 2x + 8$$

Test with $x = 1$: the original gives $5 + 3 - 2 + 7 - 1 + 1 = 13$; the answer gives $3 + 2 + 8 = 13$ ✓.

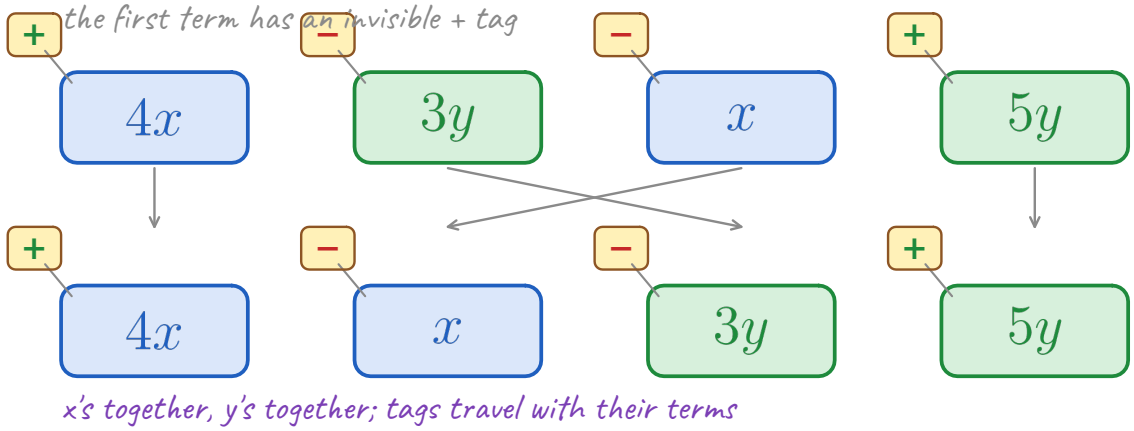
The minus sign is a luggage tag

RULE: signs belong to terms

Each term owns the sign **directly in front of it**. A first term with no sign is $+$.

You may reorder terms, but each sign moves **with** its term: $4x - 3y - x = 4x - x - 3y$.

When you sort terms, each one carries the sign **in front of it**, like a luggage tag tied to a suitcase. Move the suitcase and the tag comes too. Simplify $4x - 3y - x + 5y$:



$$4x - x - 3y + 5y \quad \longleftarrow \text{ rearranged, tags attached}$$

$$= 3x + 2y \quad \longleftarrow \text{ 4 − 1 = 3 x's; −3 + 5 = +2 y's}$$

Classic slip: reading $-3y + 5y$ as $-8y$ by "adding the 3 and 5". Use the lift from Chapter 1: start 3 floors down, go up 5 → floor 2.

Multiplying terms is different from adding them

PROCEDURE: multiplying terms

1. Multiply the **coefficients** (using the sign rules: same signs give $+$, different signs give $-$).
2. Multiply the **letters**: the same letter's powers **add** ($x \times x^2 = x^3$); different letters sit side by side ($a \times b = ab$).
3. Any terms can be multiplied; they don't need to be like terms.

To **multiply** terms you don't need like terms at all. Multiply the numbers, then the letters (using the power law $x \times x = x^2$ from Chapter 9):

$$3x \times 2x = (3 \times 2)(x \times x) = 6x^2$$ ← *numbers × numbers, letters × letters*

$$2a \times 5b = 10ab$$ ← *different letters just sit side by side*

$$-4x \times 3x^2 = -12x^3$$ ← *negative × positive = negative; $x^1 \times x^2 = x^3$*

Your Lecture 2 "common mistakes" slide, each caught by testing $x = 3$, $y = 2$:

$2x + x \neq 2x^2$	6 + 3 = 9, but 2 × 9 = 18	$2x + x = 3x$
$2x \times x \neq 3x$	6 × 3 = 18, but 3 × 3 = 9	$2x \times x = 2x^2$
$2x + 2y \neq 2xy$	6 + 4 = 10, but 2 × 6 = 12	stays $2x + 2y$
$2x + 2x^2 \neq 4x^2$	6 + 18 = 24, but 36	stays $2x + 2x^2$

The pattern: adding needs like terms and keeps the power; multiplying works on anything and adds the powers.

Expanding one bracket: everything inside gets multiplied

RULE: expanding a single bracket

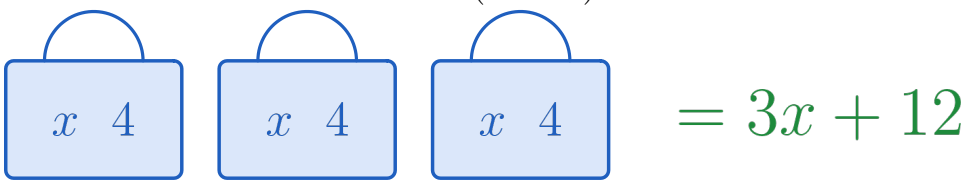
Multiply **every** term inside the bracket by the term outside, **sign included**: $a(b+c) = ab + ac$.

A minus sign in front of a bracket is -1 outside it: it **changes the sign of every term** inside. $-(x-3) = -x+3$.

Then collect like terms.

Outside	Inside	Product	Example
$+$	$+$	+	$3 \times 4 = 12$
$+$	$-$	−	$3 \times (-4) = -12$
$-$	$+$	−	$-2 \times x = -2x$
$-$	$-$	+	$-2 \times (-5) = +10$

$3(x+4)$ means three bags, each holding x and 4. Tip them all out: three x's and three 4s. So $3(x+4) = 3x + 12$.



The **grid method** does the same thing and works for everything that follows. The outside goes down the side, the inside across the top; fill each box by multiplying:

×	x	$+4$
3	$3x$	$+12$

3(x + 4) = 3x + 12

×	x	-5
-2	$-2x$	$+10$

−2 × −5 = +10 (tags multiply!)

Letters outside work the same: $x(x+3) = x^2 + 3x$.

Example: $-(x-3)$. The minus in front means $-1\times$ the bracket.

$$-1 \times x + (-1) \times (-3)$$ ← every term inside gets × (−1)

$$= -x + 3$$ ← (−1)(x) = −x; (−1)(−3) = +3

Example: $-x(2x-5)$.

$$(-x)(2x) + (-x)(-5)$$ ← the −x multiplies both terms

$$= -2x^2 + 5x$$ ← −x × 2x = −2x²; −x × −5 = +5x

Test with $x = 2$: $-2(4-5) = 2$ and $-8+10 = 2$ ✓.

Expand then collect. $3(x+2) - 2(x-1)$. The -2 (with its minus tag) multiplies **both** things in its bracket:

$$3(x+2) - 2(x-1)$$

$$= 3x + 6 - 2x + 2$$ ← −2 × x = −2x; −2 × −1 = +2

$$= x + 8$$ ← collect: 3x − 2x = x; 6 + 2 = 8

Two brackets: the grid is a rectangle's area

PROCEDURE: expanding two brackets

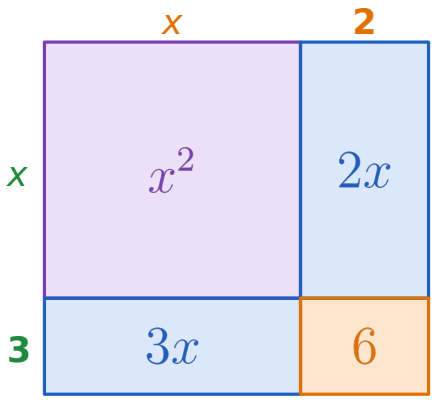
1. Multiply **each** term in the first bracket by **each** term in the second (2 × 2 = 4 products; a grid keeps track).
2. Each product takes the sign rules: watch the minus signs.
3. Collect like terms (usually the two middle x-terms).
4. Test with a number, e.g. $x = 2$, in the original and in your answer.

Your slides start with numbers: $(2+3)(4+5) = 5 \times 9 = 45$. The long way, multiplying **everything in the first bracket by everything in the second**:

$$2 \times 4 + 2 \times 5 + 3 \times 4 + 3 \times 5 = 8 + 10 + 12 + 15 = 45$$

same answer ✓

Why does that work? Draw a rectangle $2+3$ tall and $4+5$ wide and cut it into four pieces. Now do the same with letters: $(x+3)(x+2)$:



×	x	$+2$
x	x^2	$+2x$
$+3$	$+3x$	$+6$

the grid is the rectangle, tidied

$$(x+3)(x+2) = x^2 + 2x + 3x + 6 = x^2 + 5x + 6$$

collect the two blue x-terms

Test $x = 1$: $(4)(3) = 12$ and $1 + 5 + 6 = 12$ ✓. With a minus: $(x+4)(x-3)$: the grid gives $x^2 - 3x + 4x - 12 = x^2 + x - 12$ (the -3 tag makes its row of boxes negative).

Perfect squares: the two missing rectangles

RULE: three special products

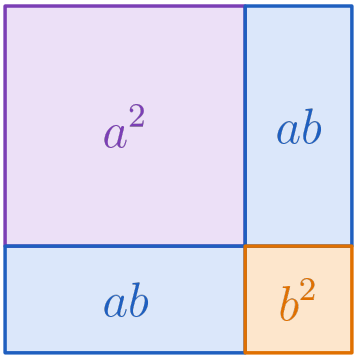
$(a+b)^2 = a^2 + 2ab + b^2$

$(a-b)^2 = a^2 - 2ab + b^2$

$(a+b)(a-b) = a^2 - b^2$ (the **difference of two squares**: the middle terms cancel)

Squaring a bracket is **not** squaring each term: $(a+b)^2 \neq a^2 + b^2$.

A very common mistake is $(a+b)^2 = a^2 + b^2$. The picture shows why it's wrong. A square of side $a+b$ has **four** pieces, not two:



$a^2 + b^2$ only counts the purple and orange squares. The two blue rectangles are forgotten, and together they're $2ab$.

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a-b)^2 = a^2 - 2ab + b^2$$

Using it (Lecture 2 example): expand $(2x-3y)^2$. Replace a by $2x$ and b by $3y$, in brackets, exactly like substitution:

$$(2x)^2 - 2(2x)(3y) + (3y)^2$$ ← *$a \to 2x$, $b \to 3y$*

$$= 4x^2 - 12xy + 9y^2$$ ← *$(2x)^2 = 4x^2$, $2 \times 2 \times 3 = 12$, $(3y)^2 = 9y^2$*

Difference of two squares. $(a-b)(a+b)$: the grid gives $a^2 + ab - ab - b^2$. The middle terms **cancel each other**, leaving $a^2 - b^2$. Chapter 12 runs this backwards.

Three brackets, and catching a slide mistake

PROCEDURE: three or more brackets

1. Expand **two** of the brackets and collect like terms.
2. Multiply that result by the next bracket: every term by every term (a bigger grid).
3. Collect like terms again, and test with $x = 1$ **and** $x = 2$.

$(x+1)^3 = (x+1)(x+1)(x+1)$. Do it in two stages: first two brackets, then multiply the result by the last one with a bigger grid:

$$(x+1)(x+1) = x^2 + 2x + 1$$ ← *stage 1: a perfect square*

$\times$	x^2	$+2x$	$+1$
x	x^3	$+2x^2$	$+x$
$+1$	$+x^2$	$+2x$	$+1$

stage 2: $(x+1) \times (x^2 + 2x + 1)$
same colours = like terms

$$(x+1)^3 = x^3 + 3x^2 + 3x + 1$$ ← *collect: $2x^2 + x^2 = 3x^2$, $x + 2x = 3x$*

Slide check. Lecture 2 writes $(x+1)^3 = x^3 + 4x^2 + 3x + 1$. Test with $x = 1$: the left side is $2^3 = 8$. The slide's version gives $1 + 4 + 3 + 1 = 9$ ✗, ours gives $1 + 3 + 3 + 1 = 8$ ✓. (The slide's own working has $2x^2 + x^2 + x^2$, one x^2 too many.)

The "MASSIVE example" $(x+4)(x+3)(x+2)(x+1)$: the slide ends with $44x^2$. With $x = 1$ the left side is $5 \times 4 \times 3 \times 2 = 120$, but $1 + 10 + 44 + 50 + 24 = 129$ ✗. The correct answer is $x^4 + 10x^3 + 35x^2 + 50x + 24$ (it sums to 120 ✓). On the slide, one "$12x^2$" was copied as "$21x^2$".

The lesson: putting a number into both sides takes ten seconds and catches most expanding mistakes, including your lecturer's. $x = 1$ is quick, but it can't tell x^2 from x^3 (both are 1), so use $x = 2$ as well when powers are involved.

Find the unknown: matching coefficients

A common question gives an expansion with a letter missing, like "$(x+a)(x+3) = x^2 + 7x + 12$: find a". The equals sign here means **identical** (true for every x, Chapter 8's $\equiv$), so each kind of term must match:

PROCEDURE: matching coefficients

1. Expand and simplify the side containing the unknown.
2. Collect it into x^2 terms, x terms and constants.
3. The two sides are identical, so the **coefficients must match**: x^2 with x^2, x with x, constant with constant. Each match is a small equation.
4. Solve one of them for the unknown, then use the others as a **check**.

Example: $(x+a)(x+3) = x^2 + 7x + 12$. Find a.

$$x^2 + 3x + ax + 3a$$ ← expand (grid: x^2, $3x$, ax, $3a$)

$$= x^2 + (a+3)x + 3a$$ ← collect the x-terms

$$x: \ a + 3 = 7 \ \Rightarrow \ a = 4$$ ← match the x coefficients

$$\text{constant: } 3a = 3(4) = 12 \ \checkmark$$ ← the constants agree too

Example: $2(x+k) - 3 = 2x + 5$. Find k.

$$2x + 2k - 3 = 2x + 5$$ ← expand the bracket

$$2k - 3 = 5 \ \Rightarrow \ k = 4$$ ← x-terms already match; match the constants

Example: $(x+p)^2 = x^2 + 10x + 25$. Find p.

$$x^2 + 2px + p^2$$ ← perfect square

$$2p = 10 \ \Rightarrow \ p = 5; \quad p^2 = 25 \ \checkmark$$ ← match x, then check the constant

Example: $3(x+2)-a(x-1)=x+8$. Find a.

$$3x+6-ax+a=(3-a)x+(6+a)$$ ← $-a \times -1 = +a$

$$3-a=1 \;\Rightarrow\; a=2; \quad 6+a=8 \;\checkmark$$

If the check fails, either you expanded wrongly or no value works. Go back to the expansion first.

Mistakes to avoid

✗ Common mistakes

✗ $-(x-3)=-x-3$ ✓ $-(x-3)=-x+3$

the minus changes the sign of EVERY term inside

✗ $3(x+4)=3x+4$ ✓ $3(x+4)=3x+12$

the 3 multiplies every term, not just the first

✗ $-x(2x-5)=-2x^2-5x$ ✓ $-2x^2+5x$

−x × −5 = +5x

✗ $(x+3)^2=x^2+9$ ✓ $(x+3)^2=x^2+6x+9$

two brackets: don't forget the 2 × 3x

✗ $\dfrac{2x+3}{x}=2+3$ ✓ $\dfrac{2x+3}{x}=2+\dfrac{3}{x}$

dividing by x divides every term; 3 has no x to cancel

✗ $2(1+3)=\dfrac{16}{x}$ ✓ $2x=13, \quad x=6.5$

Lecture 2's "dividing by x" mistake, solving 2x + 3 = 16: x was taken out of 2x only. Undo the + 3 first, then ÷ 2

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Like terms	same letters AND same powers; only the coefficient may differ
Collecting	$3x^2 - 5x^2 = -2x^2$ add the coefficients; letters and powers unchanged
Unlike terms	$x + x^2, \quad 7a + 3b$ cannot be combined by adding
Signs	each term owns the sign in front of it; the sign moves with the term
Multiplying terms	$-4x \times 3x^2 = -12x^3$ coefficients × coefficients; add powers of the same letter
Sign rules	$(+)(+) = +, \ (-)(-) = +, \ (+)(-) = -$
One bracket	$a(b + c) = ab + ac$ every term inside, sign included
Minus outside	$-(x - 3) = -x + 3$ changes the sign of every term inside
Two brackets	$(a + b)(c + d) = ac + ad + bc + bd$ each term by each term, then collect
Perfect squares	$(a \pm b)^2 = a^2 \pm 2ab + b^2$ never just $a^2 + b^2$
Difference of two squares	$(a + b)(a - b) = a^2 - b^2$
Three brackets	two at a time, collecting in between
Dividing by x	$\frac{2x + 3}{x} = 2 + \frac{3}{x}$ divide every term
Matching coefficients	in an identity, match x^2, x and constant terms to find a missing letter
Check	test with $x = 1$ and $x = 2$

Shopping bags: your turn

Answers in grey. Test each answer with x = 1 or x = 2!

Collect like terms

1. $5a + 3b - 2a + b$
2. $4x^2 - x + 3x^2 + 5x - 2$
3. $7p - 3q - 9p + q$

1. $3a + 4b$ 2. $7x^2 + 4x - 2$ 3. $-2p - 2q$

Multiply

4. $4x \times 3y$
5. $-2a \times 5a$
6. $3x^2 \times 2x^3$

4. $12xy$ 5. $-10a^2$ 6. $6x^5$

One bracket

7. $5(2x - 3)$
8. $-3(x - 4)$
9. $x(2x + 7)$
10. $2(x + 5) - 3(x - 1)$

7. $10x - 15$ 8. $-3x + 12$ 9. $2x^2 + 7x$ 10. $-x + 13$

Two brackets

11. $(x + 5)(x + 2)$
12. $(x - 3)(x + 7)$
13. $(2x + 1)(x - 4)$
14. $(x - 6)^2$
15. $(3x + 2)(3x - 2)$

11. $x^2 + 7x + 10$ 12. $x^2 + 4x - 21$ 13. $2x^2 - 7x - 4$ 14. $x^2 - 12x + 36$ 15. $9x^2 - 4$

Minus signs outside

16. $-(2x - 7)$
17. $-x(3x - 4)$
18. $4 - 2(x - 5)$

16. $-2x + 7$ 17. $-3x^2 + 4x$ 18. $-2x + 14$

Find the unknown

19. $(x + a)(x + 3) = x^2 + 7x + 12$. Find a.

20. $2(x + k) - 3 = 2x + 5$. Find k.

21. $(x + p)^2 = x^2 + 10x + 25$. Find p.

22. $a(x - 2) + 3x = 7x - 8$. Find a.

23. $(2x + 1)(x + b) = 2x^2 + 7x + 3$. Find b.

24. $5x - 3(x - c) = 2x + 12$. Find c.

19. $a = 4$ ($a + 3 = 7$; check $3a = 12$) 20. $k = 4$ ($2k - 3 = 5$) 21. $p = 5$ ($2p = 10$; $p^2 = 25$) 22. $a = 4$ ($a + 3 = 7$; $-2a = -8$) 23. $b = 3$ (constants: $b = 3$; check $2b + 1 = 7$) 24. $c = 4$ ($3c = 12$)